

Vector Space

Remember the taste test from Embeddings, where Pepsi looked almost exactly like Coke until a Citrus dimension finally told them apart? Here it is again.

Taste Profile - Coke vs. Pepsi vs. Coffee

TOKEN	TOKEN ID	DIMENSIONS						
		SWEET	BITTER	FIZZ	HEAT	CAFFEINE	DARK	CITRUS
 Coke	24317	9	1	10	2	3	8	1
 Pepsi	38106	9	1	10	2	3	8	10
 Coffee	51820	1	9	0	9	8	10	0

Notice how Coke and Pepsi’s vectors (their rows of numbers) sit much closer to each other than either does to Coffee.

This matters for AI. The model measures closeness with **distance**: how far apart two vectors are when you add up the gap in every dimension, not just a count of which slots differ. The model has established that Coke is more like Pepsi than like Coffee.



WHY THIS MATTERS

After the token runs through the layers, the new set of numbers doesn’t match a vector for any other token. It’s one of a kind.

So how does AI know what it means, when the numbers are unique? It reads its position. This is the idea of **vector space**: think of it like a map where every vector sits somewhere, and similar words share a neighborhood.

Let’s demonstrate this by using the example in the Transformer lesson.



IT starts off with a vector that’s ambiguous, and certainly doesn’t mean **CAT**. But as the token runs through the layers, its final numbers land closer to **CAT** than to any other word in the sentence.



By the last layer, meaning is a position. The model has turned your word into a point on the map: a rich, context-aware representation, not an answer yet. The next word it picks is whatever sits closest.